

Probability PhD Exam, May 2023

Carefully justify your answers.

1. Give the mathematical definition of:

1. conditional expectation,
2. martingale,
3. Poisson process,
4. standard Brownian motion.

2. This problem has three parts.

1. State the weak and strong law of large numbers for i.i.d. random variables.
2. Prove the strong law of large numbers under the additional assumption of finite fourth moment ($\mathbb{E}[X_1^4] < +\infty$).
3. Let $\{B_t\}$ be a standard Brownian motion. Prove that $\left\{\frac{B_n}{n}\right\}$ converges to 0 almost surely as $n \rightarrow +\infty$ ($n \in \mathbb{N}$).

3. This problem has two parts.

1. Let X be a random variable with a standard Gaussian distribution. Find the probability density function of e^X .
2. Let X be a random variable with a standard Cauchy distribution. Find the probability density function of $\frac{1}{X}$.

4. This problem has two parts.

1. Let $p \geq 1$. Prove that if X is a random variable, then

$$\mathbb{E}[|X|^p] = \int_0^{+\infty} px^{p-1}\mathbb{P}(|X| \geq x) dx.$$

2. Let $p \geq 1$. Let X and Y be two independent random variables with $\mathbb{E}[Y] = 0$. Show that

$$\mathbb{E}[|X + Y|^p] \geq \mathbb{E}[|X|^p].$$

5. Let $\{a_n\}_{n \geq 1}$ be a sequence of real numbers such that $0 < \sum_{k=1}^{+\infty} |a_k|^2 < +\infty$. Denote

$$\|a_n\| = \sqrt{\sum_{k=1}^n |a_k|^2} \quad \text{and} \quad \|a\| = \sqrt{\sum_{k=1}^{+\infty} |a_k|^2}.$$

Let $\{X_n\}_{n \geq 1}$ be a sequence of i.i.d. symmetric Bernoulli random variables, that is,

$$\forall k \geq 1, \quad \mathbb{P}(X_k = 1) = \mathbb{P}(X_k = -1) = \frac{1}{2}.$$

For $n \geq 1$, denote $S_n = \sum_{k=1}^n a_k X_k$, and denote $\mathcal{F}_n = \sigma\{S_1, \dots, S_n\}$.

1. Prove that $\{S_n\}$ is an $\{\mathcal{F}_n\}$ -martingale.
2. Prove that $\{S_n\}$ converges almost surely.
3. Prove that for all $n \geq 1$, for all $\lambda \in \mathbb{R}$,

$$\mathbb{E}[e^{\lambda S_n}] \leq e^{\frac{\lambda^2 \|a_n\|^2}{2}}.$$

(Hint: $\frac{e^x + e^{-x}}{2} \leq e^{x^2/2}$)

4. Use question 3 and the symmetry of S_n to prove that for all $n \geq 1$, for all $x \geq 0$,

$$\mathbb{P}(|S_n| \geq x) \leq 2e^{-\frac{x^2}{2\|a_n\|^2}}.$$

(Hint: The minimum of the function $\lambda \mapsto e^{-\lambda x} e^{\frac{\lambda^2 \|a_n\|^2}{2}}$ is attained at $\lambda = \frac{x}{\|a_n\|^2}$.)

5. Deduce the Khintchine inequality: $\forall p \geq 1, \forall n \geq 1$,

$$\mathbb{E} \left[\left| \sum_{k=1}^n a_k X_k \right|^p \right]^{\frac{1}{p}} \leq C \|a_n\|,$$

where C is a numerical constant depending on p only (you do not need to compute C).

(Hint: $\mathbb{E}[|X|^p] = \int_0^{+\infty} p x^{p-1} \mathbb{P}(|X| \geq x) dx$.)

6. This problem has three parts.

1. Give the definition of convergence of a sequence of random variables in probability and in distribution.
2. Which mode of convergence is stronger between convergence in probability and in distribution? Prove it.
3. Give a counterexample showing that convergence in probability is not equivalent to convergence in distribution.

7. Let $\{B_t\}$ be a standard Brownian motion. Define, for $x \in \mathbb{R}$,

$$T_x = \min\{t \geq 0 : B_t = x\}.$$

1. Prove that for all $x \in \mathbb{R}$, T_x is finite almost surely (that is, $\mathbb{P}(T_x < +\infty) = 1$).
2. Find $\mathbb{P}(T_{-2} < T_1)$. You can use a picture as justification.

8. Let $n \in \mathbb{N}$. Let $X_1, \dots, X_n$ be i.i.d. random variables in L_1 . Find $\mathbb{E}[X_1 \mid X_1 + \dots + X_n]$.